

Are MMALS Hosts More Than Virtual Redundancy?

Axis A Evidence, Mixed Host Ecology, and Reviewer-Driven Limits in Frozen RC2O

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Abstract

A central architectural claim in MMALS is that multiple hosts should act as functional partners rather than as redundant subnetworks that all compute the same transformation. A legitimate reviewer concern is therefore that multi-hosting may be only virtual: several apparent hosts could all lead to the same conclusions, making the architecture a descriptive metaphor rather than an evidenced mechanism. This report updates the Axis A host-specialization audit on frozen MMALS v1.1-RC2O checkpoints. The audit is strictly post-hoc: it does not alter the frozen selector, router, guard, readout, temperature, loss, or checkpoints. It combines routing concentration, host ablation, mutualistic gain, route-swap interventions, collapse controls, representation separation, and cross-seed host-profile alignment. Across MNIST, RotatedMNIST, and PermutedMNIST robust overlap-chain regimes, the audit finds measurable functional specialization in a subset of host-context pairs: 6/25, 7/25, and 6/25 respectively. This rejects the strongest null hypothesis that hosts are purely virtual or identical aggregators. It does not prove that all hosts specialize, that specialization is uniformly distributed, that the mutualism mechanism is the unique cause, or that the architecture is necessary relative to an equal-parameter baseline. The best current interpretation is a *mixed host ecology*: specialized hosts emerge where context-specific intervention matters, while generic/shared hosts remain useful for reusable transformations, stability, and efficient reuse.

1 Reviewer question and claim discipline

The reviewer question is simple and important:

Are MMALS hosts functionally specialized, or are they redundant subnetworks that only create the impression of modularity?

This report treats the question as a causal and diagnostic problem. Routing specialization alone is not enough. A host can receive high route mass and still be a cosmetic endpoint. Conversely, a host can be generic and useful without being context-specialized. We therefore require agreement between several independent signals.

Supported claim. Under frozen RC2O checkpoints, MMALS rejects the strongest “virtual host” null: several host-context pairs are not interchangeable, because removing or swapping the routed host can change accuracy, cross-entropy, output distribution, or latent state. The supported empirical claim is that frozen RC2O exhibits *measurable functional host specialization in a subset of host-context pairs*.

Unsupported claims. Axis A does not prove biological specialization, universal host specialization, production readiness, or causal sufficiency of the MMALS mutualism mechanism. It does not eliminate the capacity confound without equal-parameter controls. It also does not imply that generic hosts are bad. A mixed structure can be the correct architecture when a system must share reusable transformations while still assigning specialized partners to brittle contexts.

Table 1: Reviewer-facing status after Axis A. Axis A addresses the strongest virtual-host null, but it does not close every architectural criticism.

Critique	Status after Axis A	Honest interpretation
Hosts are identical aggregators	Addressed empirically	Ablation, route swap, collapse, and CKA contradict pure virtual redundancy
Specialization ratio is modest	Expected but needs framing	6-7/25 pairs means mixed ecology, not universal specialization
One host dominates	Important vulnerability	Evidence suggests hub-and-partners, not evenly distributed multi-hosting
Equal-parameter control absent	Open and critical	Must be tested before claiming architectural necessity
RotatedMNIST cross-seed profile stability weak	Open empirical warning	Supports non-virtual function, not stable host taxonomy
Mutualism causes specialization	Not proven by Axis A	Axis A audits consequences; Axis B must test mechanism

Table 2: Dataset-level Axis A evidence summary. Final accuracy and forgetting are robust five-seed RC2O metrics from the freeze manifests. Route dominance, CKA separation, and claim counts are computed from the Axis A seed-0 causal audit; cross-seed scores use aligned host profiles over seeds 0–4.

Dataset	Final acc.	Min-task	Forgetting	Route dom.	CKA sep.	Func. pairs	Xseed cos.	Xseed corr.
MNIST	0.965	0.950	0.020	0.809	0.734	6/25	0.978	0.928
RotatedMNIST	0.968	0.948	0.021	0.840	0.753	7/25	0.961	0.184
PermutedMNIST	0.970	0.940	0.022	0.792	0.645	6/25	0.920	0.513

2 Background and scope

MMALS is a continual-learning architecture with host modules, a routing medium, inferred context, memory, and audit layers. Architecturally, this places MMALS near mixtures of local experts [Jacobs et al., 1991], sparsely gated expert systems [Shazeer et al., 2017, Fedus et al., 2021], modular continual-learning systems [Rusu et al., 2016], and representation-similarity analysis such as linear CKA [Kornblith et al., 2019]. The present report is narrower than the full MMALS thesis: it asks whether the hosts in frozen RC2O artifacts are observably non-virtual.

Consequence audit, not mechanism proof. Axis A audits the consequence of the architecture: do hosts behave as distinguishable, context-sensitive functional contributors? It does not yet prove why that specialization emerged. In particular, it does not prove that the proposed mutualistic route-function-memory mechanism is the unique causal source of specialization. That requires an additional mechanism-ablation axis.

3 Frozen candidate and datasets

The audit uses MMALS v1.1-RC2O frozen checkpoints. RC2O is treated as read-only. No selector, router, guard, head policy, temperature, or loss coefficient is changed during Axis A. The checkpoint identity rule is `dataset + regime + seed + selected policy + config`. The evaluated regimes are MNIST overlap chain, RotatedMNIST with rotations $[-30, -15, 0, 15, 30]$, and PermutedMNIST with permutation seeds [100, 101, 102, 103, 104]. Each robust package contains five seed checkpoints.

4 Decision rule: what counts as functional specialization?

A host-context pair is not declared specialized because of a single number. The rule-based classifier combines routing specialization, host usefulness, functional contribution, representation separation, non-triviality, and stability. Host usefulness is measured through positive mutualistic gain,

$$G_i(c) = \mathcal{L}_{\text{without } i}(c) - \mathcal{L}_{\text{full}}(c). \quad (1)$$

Table 3: Claim counts from the conservative rule-based classifier. Counts are host-context pairs out of 25 in the seed-0 audit.

Dataset	Functional	Routing-only	No claim
MNIST	6	1	18
RotatedMNIST	7	2	16
PermutedMNIST	6	1	18

Table 4: Decision rule used to count functional host-specialization pairs. The Host Specialization Index ranks pairs but does not decide the claim by itself.

Component	Rule / threshold	Role in claim
Routing specialization	mean entropy $< 0.70 \log(H)$ and mean dominance > 0.45	Router is not broadcasting uniformly.
Meaningful host routing	route weight $> 0.75/H$ or host is dominant	Candidate host is actually used.
Positive mutualistic gain	$G_i(c) = \mathcal{L}_{\text{without } i}(c) - \mathcal{L}_{\text{full}}(c) > 10^{-4}$	Removing the host worsens local objective.
Ablation sensitivity	accuracy drop > 0.01 or CE increase > 0.01 or output KL $> 10^{-3}$ or latent MSE $> 10^{-4}$	Host has causal behavioral or latent effect.
Representation separation	CKA separation > 0.05	Host transformations are not identical copies.
Functional-specialization label	all five conditions above are true	Conservative rule-based claim.

The exact rule is intentionally placed in the main text because the claim counts depend on it. Functional host specialization requires all of the following: route specialization at context level, meaningful host routing, positive mutualistic gain, ablation sensitivity, and representation separation. The Host Specialization Index (HSI) is only a ranking score; it is not used alone to assign the claim.

5 Main empirical result: partial, not universal specialization

Figure 1 summarizes the conservative claim counts. Only a subset of host-context pairs crosses the full rule. Most pairs are not claimed to be specialized.

Table 5: Highest-scoring functional host-specialization pairs in each dataset. Gain is cross-entropy without the host minus cross-entropy with the full system. A positive gain and positive accuracy drop indicate causal usefulness under ablation.

Dataset	Task	Host	Route	Gain	Acc. drop	Output KL	HSI
MNIST	2 vs 3	H1	0.996	9.296	0.459	8.080	0.952
MNIST	4 vs 5	H1	0.980	7.423	0.546	5.950	0.891
MNIST	3 vs 4	H1	0.667	1.373	0.282	1.346	0.455
MNIST	1 vs 2	H1	0.565	0.130	0.041	0.315	0.277
RotatedMNIST	2 vs 3 @ +0deg	H1	0.977	10.785	0.470	7.681	0.988
RotatedMNIST	1 vs 2 @ -15deg	H1	0.538	1.809	0.474	1.544	0.537
RotatedMNIST	4 vs 5 @ +30deg	H1	0.907	1.389	0.122	1.310	0.464
RotatedMNIST	3 vs 4 @ +15deg	H1	0.850	0.133	0.021	0.269	0.363
PermutedMNIST	3 vs 4 @ perm_seed=103	H3	0.688	5.364	0.485	5.725	0.860
PermutedMNIST	2 vs 3 @ perm_seed=102	H3	0.998	2.897	0.331	1.640	0.679
PermutedMNIST	3 vs 4 @ perm_seed=103	H1	0.311	3.320	0.485	2.918	0.598
PermutedMNIST	4 vs 5 @ perm_seed=104	H1	0.588	1.741	0.475	1.887	0.583

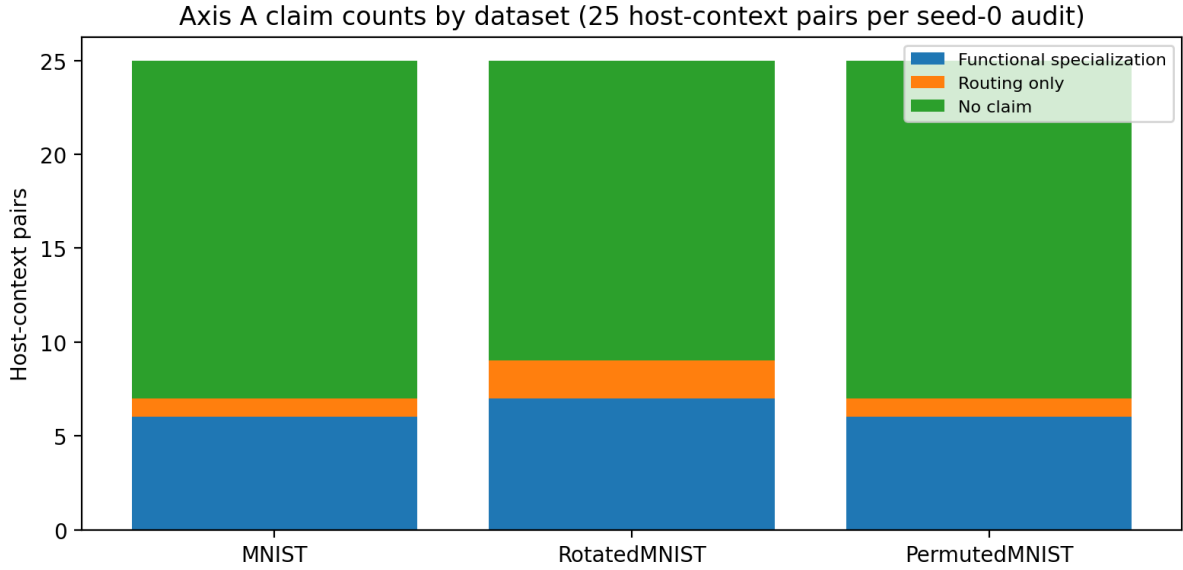


Figure 1: Rule-based Axis A claim counts. Each dataset has 25 host-context pairs in the seed-0 causal audit. Functional specialization is a minority pattern, not a universal label.

MNIST, RotatedMNIST, and PermutedMNIST all show non-zero functional specialization: 6, 7, and 6 pairs respectively. This is neither “all hosts are the same” nor “every host is a specialist.” It is a mixed host ecology. A hostile reading could say that 6–7/25 is a modest ratio. The more precise response is that 25/25 would not necessarily be desirable: it could indicate over-fragmentation, poor reuse, higher cost, and brittle specialization. The right target is not universal specialization; it is specialization where causal intervention shows that specialization matters.

Table 6: Observed host ecology. The evidence does not show equal participation of all hosts. It shows a hub-and-partners pattern: one specialist/hub often carries most differentiated work, while auxiliary and generic hosts remain useful in some contexts.

Dataset	Observed pattern	Empirical signal	Interpretation
MNIST	H1 hub; H4 satellite	5 of 6 seed-0 functional pairs on H1; 31 functional pairs over five seeds before alignment.	Clean overlap-chain regime: mostly a single specialist/hub plus limited satellite specialization.
RotatedMNIST	H1 hub; H4 early-rotation auxiliary	5 of 7 seed-0 pairs on H1; 2 on H4; cross-seed profile correlation mean = 0.184.	Geometry-shift regime: non-virtual host roles exist, but the exact host taxonomy is not stable across seeds.
PermutedMNIST	H3 domain-shift hub; H1 late-context partner	4 of 6 seed-0 pairs on H3; 2 on H1; strongest single-global collapse drop = 0.375.	Permutation regime: clearer hub-plus-partner behavior; not uniform multi-host democracy.

6 Host dominance: multi-hosting is not evenly distributed

A second reviewer concern is that specialization may collapse to one hub host with generic satellites. The data partly supports this concern and should be stated explicitly. MNIST and RotatedMNIST concentrate most of their highest-scoring functional pairs on H1, while PermutedMNIST concentrates them mostly on H3 with H1 as a later-context partner.

This does not invalidate the host-specialization claim, but it narrows it. The current evidence supports a *hub-and-partners ecology*, not a uniform multi-host democracy. The single-global-host collapse control is therefore important: if one host were sufficient, forcing all contexts through the best global host should not damage performance. In practice, the single-global-host control produces positive accuracy drops, especially in PermutedMNIST, but a stronger pruned single-hub experiment is still required.

7 Routing evidence

Route concentration asks whether the model broadcasts to all hosts or selects hosts differently across contexts.

Mean route by context/task across Axis A datasets

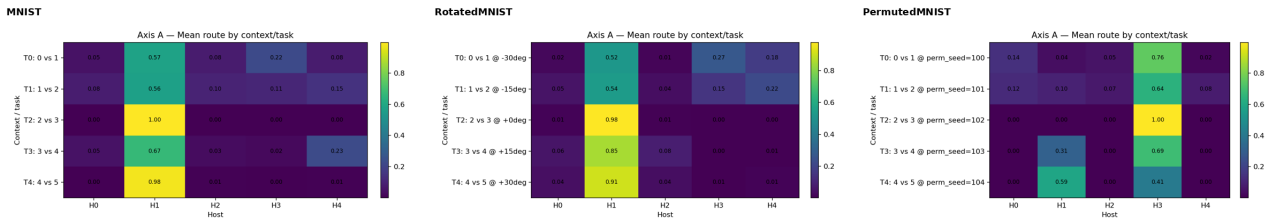


Figure 2: Mean route by context/task across datasets. Routing is not uniform broadcasting. However, routing alone is not treated as proof of specialization.

The heatmaps show non-uniform route allocation. However, the distribution often has a hub structure. Therefore the claim is not that all hosts are equally involved, but that host-route assignments are sometimes causally meaningful.

8 Causal intervention evidence: ablation and route swap

Host ablation removes one host, renormalizes the route over the remaining hosts, and measures the consequence. If a supposedly useful host is virtual, removing it should not change behavior much.

Host ablation accuracy drop across Axis A datasets

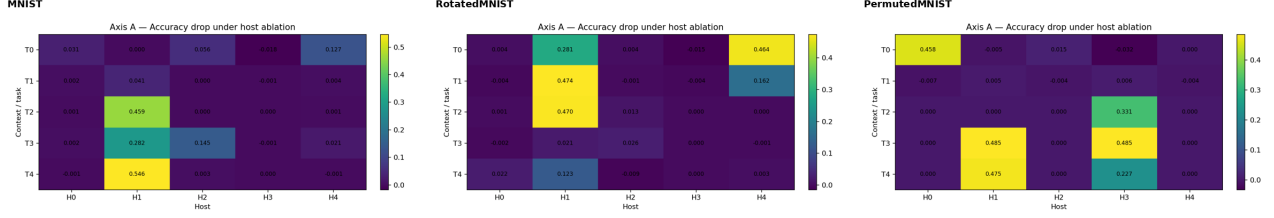


Figure 3: Host ablation accuracy drop. Large drops indicate that specific hosts have causal contribution in specific contexts. Negative or near-zero drops identify generic, redundant, or non-useful host-context pairs.

Ablation is the strongest evidence against pure virtual redundancy: some host removals cause large accuracy drops and CE increases in specific contexts. The result is not uniform; many removals do not matter. That is exactly why the article does not claim universal specialization.

Route swaps evaluate a task using the mean route of another task. This tests whether the route-function binding is context-specific.

Route swap accuracy drop across Axis A datasets

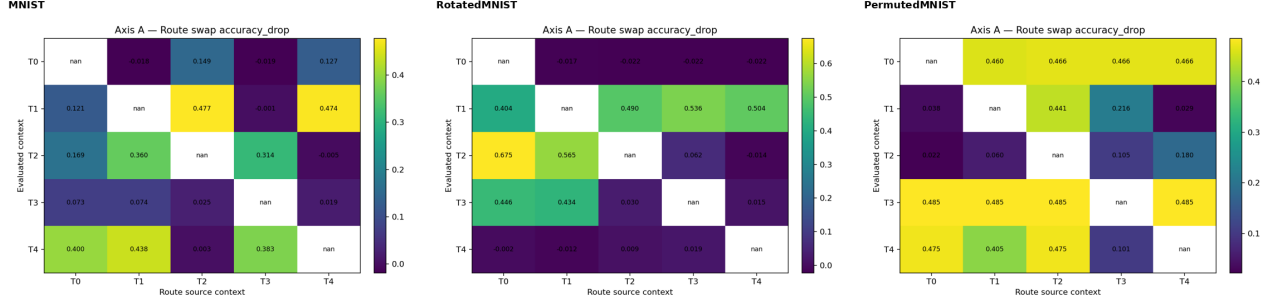


Figure 4: Route-swap accuracy drop. Swapping routes often damages task performance, supporting non-arbitrary route-function binding. The effect is not uniform, which is consistent with a mixture of shared and specialized routes.

9 Non-triviality controls and the equal-parameter gap

Collapse controls test whether the observed behavior is explained by post-hoc route labeling rather than by learned route-host structure. Uniform routing forces every host to be used equally. Shuffled routing reassigns routes to wrong contexts. Single-global-host routing forces all contexts through one host.

Table 7: Mean accuracy drop under non-triviality controls. A positive drop means that forcing the host system to behave less selectively damages performance.

Dataset	Uniform route	Shuffled route	Single global host
MNIST	0.269	0.137	0.132
RotatedMNIST	0.217	0.077	0.105
PermutedMNIST	0.235	0.279	0.375

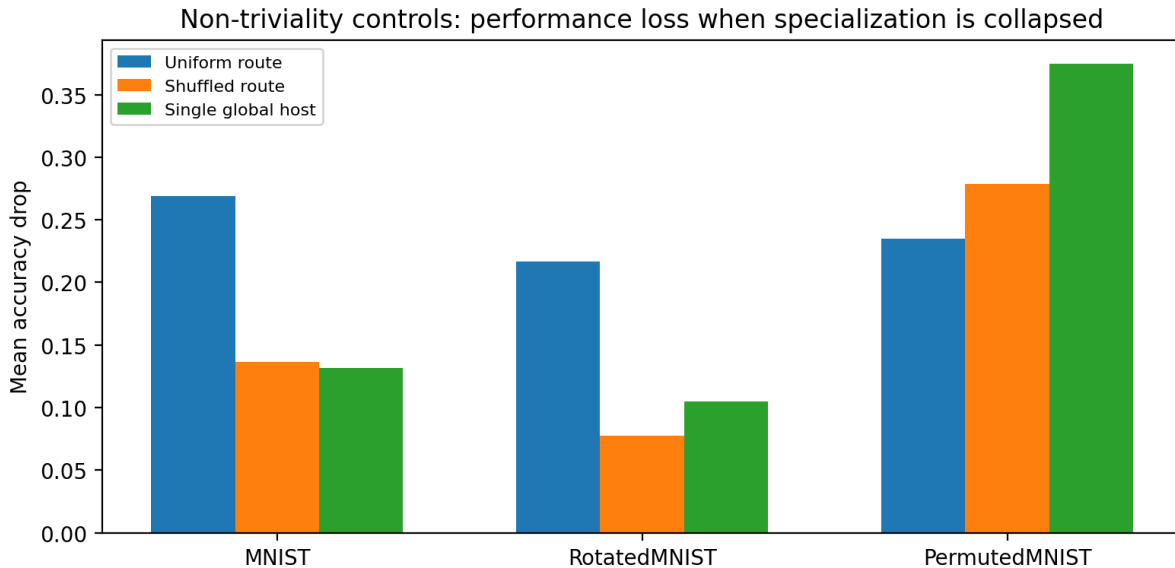


Figure 5: Mean accuracy drop under collapse controls. Positive drops indicate that the learned route-host structure matters beyond merely having multiple subnetworks.

These controls are useful, but they are not the same as an equal-parameter baseline. The current article can claim that learned host-route structure is not cosmetic under the tested interventions. It cannot yet claim that MMALS is architecturally necessary compared with an equal-parameter monolith or a standard MoE. This is the most important remaining control for a MoE-aware reviewer.

10 Representation separation

Representation separation uses linear CKA between transformed host features. If hosts were identical copies, off-diagonal CKA would be high and separation would be low. The mean CKA separation is 0.734 for MNIST, 0.753 for RotatedMNIST, and 0.645 for PermutedMNIST. This supports host heterogeneity, but by itself it is not causal proof.

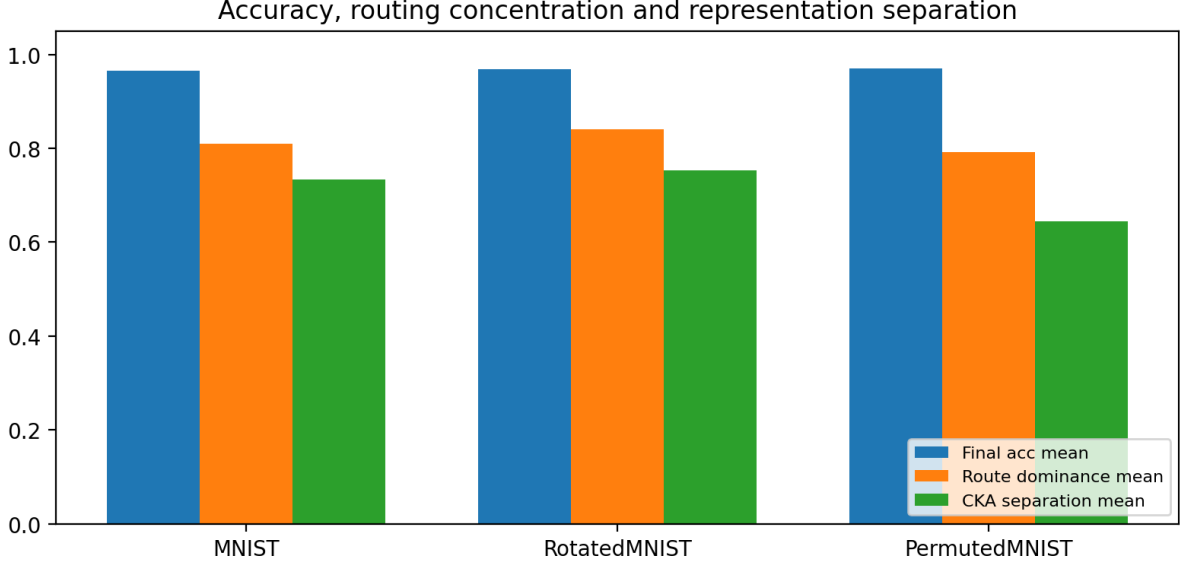


Figure 6: Accuracy, routing concentration, and representation separation. The system keeps high accuracy while using non-uniform routes and separated host transformations.

11 Cross-seed stability: strong in MNIST, unstable taxonomy in RotatedMNIST

Host IDs may permute across seeds. Therefore, the audit aligns host profiles before judging stability. The meaningful question is not whether H1 always means the same raw index, but whether equivalent host functions recur after label alignment.

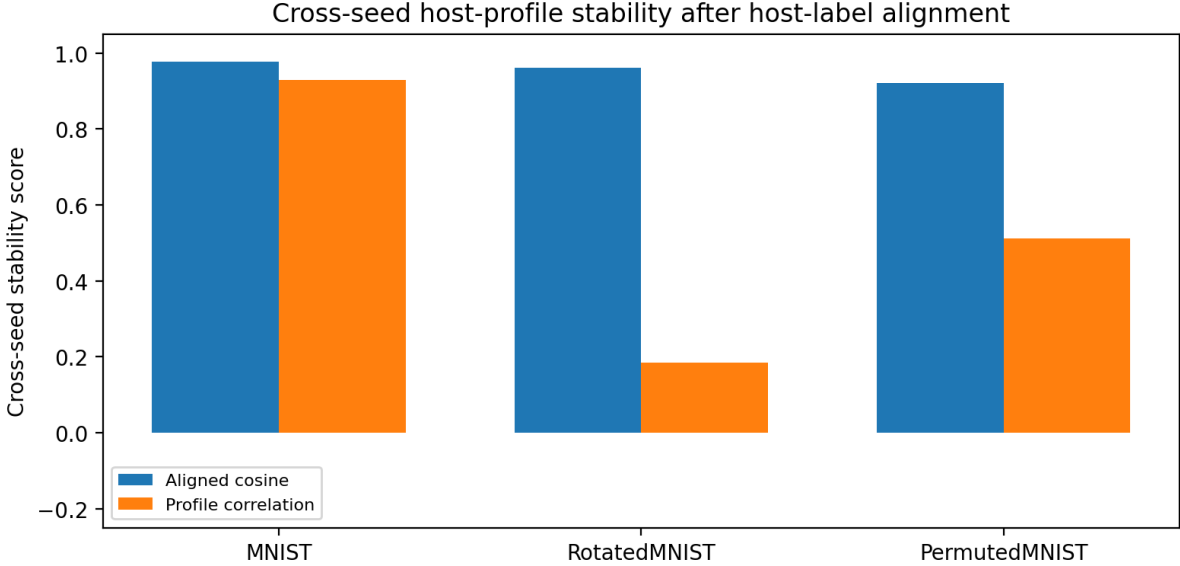


Figure 7: Cross-seed stability after host-label alignment. Mean aligned cosine is high in all three datasets. Profile correlation is strongest for MNIST, weaker for PermutedMNIST, and very weak on average for RotatedMNIST.

The RotatedMNIST result is the most serious empirical warning. Its mean aligned cosine is high (0.961), but mean flattened profile correlation is only 0.184. This means that hosts are not purely virtual, but the exact host-function profile is not stable as a taxonomy. A plausible interpretation is that continuous geometric transformations permit several equivalent decompositions: after alignment, there is broad functional similarity, but not the same context-by-host profile. Therefore RotatedMNIST supports non-virtual host function, not stable host taxonomy.

12 Are generic hosts bad?

No. The audit explicitly distinguishes specialized, useful, generic, routing-only, and no-claim host-context pairs. A generic host can be valuable when it provides a reusable transformation shared across contexts. In continual learning, a fully specialized-only architecture may overfit, consume more memory, fragment evidence, and become brittle under small shifts. Generic or shared hosts are desirable when:

- the contexts share low-level features or invariants;
- the system is data-limited and should reuse transformations;
- the downstream task benefits from stability more than from separation;
- energy or memory cost penalizes unnecessary context fragmentation;
- multiple route decompositions are functionally equivalent.

Specialized hosts are desirable when:

- a context has high interference with previous contexts;
- route swaps or ablations show large context-specific damage;
- a boundary is weak or ambiguous and needs a dedicated transformation;
- a domain shift creates a transformation that cannot be cheaply shared.

The observed result is therefore not a failure because it is partial. It is a positive but bounded result: MMALS has specialized hosts where they matter, but it also keeps generic/shared hosts where reuse is useful or where the audit has insufficient evidence to claim specialization.

13 What Axis A proves, and what it does not prove

Axis A proves less than a maximal architecture claim, but more than a visual intuition. It shows that the strongest virtual-host null is not supported by the current evidence. If hosts were only identical aggregators, ablation, swap, collapse, and CKA separation should not jointly produce the observed pattern.

However, Axis A does not prove that the observed specialization is caused specifically by MMALS mutualism. A standard MoE, an equal-parameter modular network, or a single-hub-plus-satellites design might reproduce part of the result. This is why the next axis must compare mechanism and capacity directly.

Table 8: Proposed Axis A+ controls required to move from non-virtual host evidence to stronger architectural-necessity evidence.

Control	Question answered	Expected reviewer value
Equal-parameter monolith	Is the gain just total capacity?	Directly controls the most important MoE-style confound
Equal-parameter standard MoE	Is MMALS just a generic routed expert system?	Separates MMALS route-memory structure from ordinary MoE routing
Shared-host-weights control	Do distinct host functions matter?	Tests whether routes alone explain the effect
Frozen-random-routing control	Does learned routing matter?	Tests whether specialization is post-hoc or learned
Single-hub pruned MMALS	Can one specialist plus generic readout explain everything?	Tests hub dominance and whether satellites are necessary
Mutualism ablation	Does the proposed mutualistic mechanism cause specialization?	Moves from consequence audit to causal mechanism audit

14 Reviewer-facing answer

The concise answer to the reviewer is:

Axis A does not claim that all MMALS hosts specialize. It rejects the stronger null that the hosts are merely virtual and interchangeable. In frozen RC2O, a minority but non-zero subset of host-context pairs shows causal and representational evidence of specialization. The most accurate interpretation is mixed host ecology: one or more specialist/hub hosts, auxiliary partners, and generic/shared hosts. This is useful but incomplete evidence; equal-parameter and mechanism-ablation controls remain necessary before claiming architectural necessity.

15 Limitations

The strongest limitations are explicit:

1. The specialization ratio is modest: 6–7/25 seed-0 host-context pairs pass the full rule.
2. Specialization is not evenly distributed; H1 and H3 dominate many of the strongest pairs.
3. Equal-parameter controls are absent and are critical for claims of architectural necessity.
4. RotatedMNIST cross-seed profile correlation is weak, suggesting unstable host taxonomy under continuous geometric shift.
5. Axis A audits behavioral consequence, not the causal mechanism that produced specialization.
6. The evaluated regimes are still small vision benchmarks and do not establish production readiness.

16 Conclusion

Axis A turns the host-specialization question from an architectural intuition into an auditable claim. MMALS hosts are not proven biological specialists, and they are not all specialized. But they are also not merely virtual redundant subnetworks. Across three robust frozen RC2O regimes, a subset of host-context pairs shows convergent evidence of functional specialization: concentrated routing, positive mutualistic gain, causal ablation sensitivity, route-swap degradation, collapse-control degradation, representation separation, and cross-seed alignment. The current best claim is that frozen RC2O exhibits a mixed host ecology: specialized hosts emerge for some contexts, while generic/shared hosts remain useful for reusable transformations and robustness. The next required step is Axis A+: equal-parameter, MoE, single-hub, and mutualism-ablation controls.

References

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A Artifact manifest

The GitHub package includes raw manifests, CSV tables, PNG plots, and scripts used to aggregate the results. The source Axis A manifests record checkpoint SHA-256 hashes, dataset-regime keys, config hashes, seeds, and thresholds.

B Specialization decision rule

The functional-specialization label requires route specialization at the context level, meaningful host routing, positive mutualistic gain, ablation sensitivity, and representation separation. HSI is used for ranking, not as the sole claim criterion. The exact thresholds are provided in Table 4.